

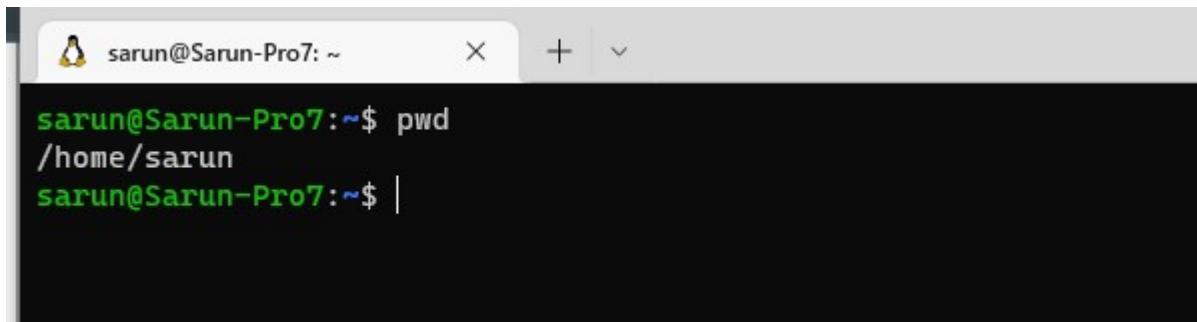
Operating Systems

Lab 1

For each question, you need to take a screenshot of the Linux commands you use and the results of the commands and put the screenshot under the question. You may use this link, <https://kb.iu.edu/d/afsk>, to find the Linux commands you may need to use.

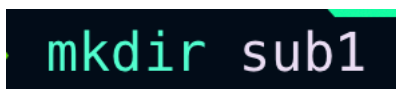
1. Show the current working directory.

(Example output)

A terminal window titled 'sarun@Sarun-Pro7: ~' with standard window controls. The terminal shows a green prompt 'sarun@Sarun-Pro7:~\$' followed by the command 'pwd'. The output is '/home/sarun' on the next line. A second green prompt 'sarun@Sarun-Pro7:~\$' is shown with a cursor, indicating the command has been executed.

```
sarun@Sarun-Pro7:~$ pwd
/home/sarun
sarun@Sarun-Pro7:~$ |
```

2. Create a subdirectory “sub1” under your home directory.

A close-up of a terminal line showing the command 'mkdir sub1' in green text on a black background.

```
mkdir sub1
```

3. Change to subdirectory you created in the previous question.

```
> cd sub1
```

4. Create a text file using the following command `cat > test1.txt`. Then enter a few lines of text then press `ctrl-d` to save the file.

```
> cat > test1.txt  
iloveyou  
youlookcool
```

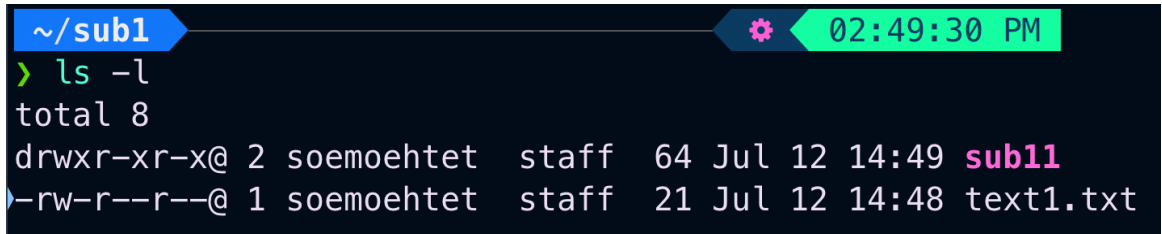
5. Create a subfolder `sub11` under your current directory.

```
~/sub1  
> mkdir sub11
```

6. List the files and folders in your working directory.

```
~/sub1  
> ls  
sub11      text1.txt
```

7. List the files and folders in your working directory by showing properties such as permissions, owners, and sizes.



```
~/sub1 02:49:30 PM
> ls -l
total 8
drwxr-xr-x@ 2 soemoetet  staff  64 Jul 12 14:49 sub11
-rw-r--r--@ 1 soemoetet  staff  21 Jul 12 14:48 text1.txt
```

8. How can we tell the difference between a file and a folder from the result of the command in the previous question? (For this question, you may give only your explanation without any screenshots)

how to distinguish between files and folders using the output of the `ls -l` command in Linux:

File type: The first character in the output indicates the file type.

For regular files (containing data or text), it will be a hyphen (-).

For directories (folders used to organize files), it will be a lowercase 'd'.

Size: Files will have a size listed in bytes, while folders will show a total size of 0 bytes since they don't store data themselves, only references to the files they contain.

9. List the files and folders in your working directory including the hidden files.

```
~/sub1 02:49:55 PM
> ls -la
total 8
drwxr-xr-x@ 4 soemoetet staff 128 Jul 12 14:49 .
drwxr-x---@ 81 soemoetet staff 2592 Jul 12 14:50 ..
drwxr-xr-x@ 2 soemoetet staff 64 Jul 12 14:49 sub11
-rw-r--r--@ 1 soemoetet staff 21 Jul 12 14:48 text1.txt
```

10. Copy the contents of the test1.txt to the new file test2.txt.

```
~/sub1
> cat text1.txt > text2.txt

~/sub1
> ls
sub11 test2.txt text1.txt text2.txt
```

11. Copy the contents of the test1.txt to the new file test3.txt.

```
~/sub1  
> cat text1.txt > text3.txt  
~/sub1  
> ls  
sub11      test2.txt text1.txt text2.txt te  
~/sub1  
> rm text2.txt  
~/sub1  
> ls  
sub11      test2.txt text1.txt text3.txt
```

12. Rename test2.txt to test2.dat.

```
~/sub1  
> mv test2.txt test2.dat  
~/sub1  
> ls  
sub11      test2.dat text1.txt text3.txt
```

13. Delete the file test3.txt.

```
~/sub1  
> rm test3.txt  
  
~/sub1  
> ls  
sub11      test2.dat  text1.txt
```

14. Delete all files in the folder sub1.

```
~  
> rm -f sub1/*  
zsh: sure you want to delete the only file in /Users/soemoehtet/sub1 [yn]? y  
rm: sub1/sub11: is a directory
```

```
~/sub1  
> ls  
sub11
```

15. Go back to your home folder.

16. Create a folder sub12 under sub1 without changing to sub1 first.

```
> cd ..  
~  
> mkdir sub1/sub12  
~  
> cd sub1  
~/sub1  
> ls  
sub11 sub12  
~/sub1
```

17. Go to folder sub1

```
> cd ..  
~  
> mkdir sub1/sub12  
~  
> cd sub1  
~/sub1  
> ls  
sub11 sub12  
~/sub1
```

18. Create the following files in the folder sub1, file1.txt, file2.dat, file3.doc, file4.txt, file5.txt, file11.txt, and file21.dat. (Hint: The easiest way to do this is to use the “touch” command. However, if you do not know how to use “touch,” you can use a text editor to create those files or create one file and copy the contents to the rest. You do not need to take a screenshot or answer anything for this question.)
19. Copy only the files whose names begin with “file” and follow by only one digit from folder sub1, i.e., copy only file1.txt, file2.dat, file3.doc, file4.txt, and file5.txt to folder sub11. You need to use only one cp command.


```
> cp sub1/file[1-5].* sub11/
cp: sub11 is not a directory
~
> cp sub1/file[1-5].* sub1/sub11/
~
> cd sub1/sub11
~/sub1/sub11
> ls
file1.txt file2.dat file3.doc file4.txt file5.txt
```

20. List the files in folder sub11 without changing to the folder sub11 first.

```
~/sub1
> ls sub11/*
sub11/file1.txt sub11/file2.dat sub11/file3.doc sub11/file4.txt sub11/file5.txt
~/sub1
```

21. Show the manual of the ls command.

```

LS(1)                                     General Commands Manual      LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [-@ABCFGHILOPRSTUWabcdeghiklmnopqrstuvwxy1%,] [--color=when] [-D format] [file ...]

DESCRIPTION
    For each operand that names a file of a type other than directory, ls displays its name as well as any requested, associated information. For each operand that names a file of type directory, ls displays the names of files contained within that directory, as well as any requested, associated information.

    If no operands are given, the contents of the current directory are displayed. If more than one operand is given, non-directory operands are displayed first; directory and non-directory operands are sorted separately and in lexicographical order.

    The following options are available:

    -@      Display extended attribute keys and sizes in long (-l) output.

    -A      Include directory entries whose names begin with a dot ('.') except for . and .. . Automatically set for the super-user unless -I is specified.

    -B      Force printing of non-printable characters (as defined by ctype(3) and current locale settings) in file names as \xxx, where xxx is the numeric value of the character in octal. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

    -C      Force multi-column output; this is the default when output is to a terminal.

    -D format
        When printing in the long (-l) format, use format to format the date and time output. The argument format is a string used by strftime(3). Depending on the choice of format string, this may result in a different number of columns in the output. This option overrides the -T option. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

    -F      Display a slash ('/') immediately after each pathname that is a directory, an asterisk ('*') after each that is executable, an at sign ('@') after each symbolic link, an equals sign ('=') after each socket, a percent sign (%) after each whiteout, and a vertical bar (|) after each that is a FIFO.

    -G      Enable colorized output. This option is equivalent to defining CLICOLOR or COLORTERM in the environment and setting --color=auto. (See below.) This functionality can be compiled out by removing the definition of COLORLS. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

    -H      Symbolic links on the command line are followed. This option is assumed if none of the -F, -d, or -l options are specified.

    -I      Prevent -A from being automatically set for the super-user. This option is not defined in IEEE Std 1003.1-2008 ("POSIX.1").

    -L      Follow all symbolic links to final target and list the file or directory the link references rather than the link itself. This option cancels the -P option.

    -O      Include the file flags in a long (-l) output. This option is incompatible with IEEE Std 1003.1-2008 ("POSIX.1"). See chflags(1) for a list of file flags and their meanings.

    -P      If argument is a symbolic link, list the link itself rather than the object the link references.

```

22. Show the list of files in folder sub1 and you sort the result based on the extension in the ascending order.

```

~
> ls -l sub1 | sort -k 1,1 -t .
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file1.txt
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file11.txt
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file2.dat
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file21.dat
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file3.doc
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file4.txt
-rw-r--r--@ 1 soemoetet staff 0 Jul 12 15:34 file5.txt
drwxr-xr-x@ 2 soemoetet staff 64 Jul 12 15:32 sub12
drwxr-xr-x@ 7 soemoetet staff 224 Jul 12 15:37 sub11
total 0

```

23. Remove the folder sub12.

```

~/sub1
> ls
file1.txt file11.txt file2.dat file21.dat file3.doc file4.txt file5.txt sub11
~/sub1

```

24. Remove all files in folder sub11 using only one command.

```

> ls
file1.txt file11.txt file2.dat file21.dat file3.doc file4.txt file5.txt sub11
~/sub1
> rm sub11/*
zsh: sure you want to delete all 5 files in /Users/soemoetet/sub1/sub11 [yn]? y
~/sub1
> cd sub11
~/sub1/sub11
> ls

```

25. Go back to your home directory.

```
~/sub1
> cd sub11
~/sub1/sub11
> ls
~/sub1/sub11
> cd ..
~/sub1
> cd ..
~
```

26. Create folder sub11 under sub1 without going into sub1 first.

```
~
> mkdir sub1/sub11
mkdir: sub1/sub11: File exists
```

27. Copy the file file1.txt from folder sub1 to folder sub1/sub11 without going to sub1 first. **You need to refer to the file and folder using relative path.**

```
> mkdir sub1/sub11
mkdir: sub1/sub11: File exists

~
> cp sub1/file1.txt sub1/sub11/

~
> cd sub1/sub11

~/sub1/sub11
> ls
file1.txt
```

28. Copy the file file2.dat from folder sub1 to folder sub1/sub11 without going to sub1 first. **You need to refer to the file and folder using absolute path.**

```
~/sub1/sub11
> cp ~/sub1/file2.dat ~/sub1/sub11/

~/sub1/sub11
>
```

29. Remove the folder sub1 and everything inside it with one command and without user confirmation.

```
~/sub1/sub11 03:4
> rm -rf sub1/

~/sub1/sub11 03:4
> rm -rf ~/sub1/

~/sub1/sub11 03:4
> cd ..

~/sub1/sub11 03:4
> cd ..

~ 03:4
> ls
Applications      Flutter projects  Pictures          if all([True, True, True]):.py
Cisco Packet Tracer 8.2.1  Flux             Postman          import requests.py
Desktop           Fluxcommon       Public           node_modules
Dev_projects      Frontware        Untitled Notebook (6).pdf package-lock.json
Developer         KMITL           Web_projects    package.json
Documents         Library          Youtube         pouwmarketplace
Downloads         Movies          books
Flutter practice  Music           go
```